

Blockchain Experiment 11

AIM: To implement a Hyperledger Fabric network, deploy chaincode, and perform asset creation, querying, and transfer operations.

PROCEDURE

Step 1: Update the system

Go to <https://shell.cloud.google.com> and run the following commands:

```
sudo apt update
```

```
sudo apt upgrade -y
```

```
g2022_aditi_taksale@cloudshell:~$ sudo apt update
Get:1 https://cli.github.com/packages stable InRelease [3,917 B]
Get:2 https://packages.cloud.google.com/apt gcsfuse-noble InRelease [1,227 B]
Get:3 https://packages.cloud.google.com/apt cloud-sdk InRelease [1,620 B]
Get:4 https://apt.postgresql.org/pub/repos/apt noble-pgdg InRelease [180 kB]
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Get:6 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:7 https://packages.cloud.google.com/apt gcsfuse-noble/main amd64 Packages [66.1 kB]
Get:8 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0 InRelease [3,005 B]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:10 https://packages.cloud.google.com/apt cloud-sdk/main all Packages [1,982 kB]
Get:11 https://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 Packages [675 kB]
Get:12 https://packages.cloud.google.com/apt cloud-sdk/main amd64 Packages [4,640 kB]
Hit:13 https://ppa.launchpadcontent.net/dotnet/backports/ubuntu noble InRelease
Get:14 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0/multiverse amd64 Packages [68.7 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [3,505 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2,377 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3,692 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,989 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2,155 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,508 kB]
```

```
g2022_aditi_taksale@cloudshell:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  apparmor libapparmor1
The following packages will be upgraded:
  docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin google-cloud-cli
  google-cloud-cli-anthoscli google-cloud-cli-app-engine-go google-cloud-cli-app-engine-java google-cloud-cli-app-engine-python
  google-cloud-cli-app-engine-python-extras google-cloud-cli-bigtable-emulator google-cloud-cli-cbt google-cloud-cli-cloud-run-proxy
  google-cloud-cli-datastore-emulator google-cloud-cli-gke-gcloud-auth-plugin google-cloud-cli-istioctl google-cloud-cli-kpt
  google-cloud-cli-local-extract google-cloud-cli-minikube google-cloud-cli-nomos google-cloud-cli-package-go-module
  google-cloud-cli-pubsub-emulator google-cloud-cli-scaffold google-cloud-cli-spanner-cli kubect1 libarchive13t64 libgdk-pixbuf-2.0-0
  libgdk-pixbuf2.0-common libpam-systemd libpq-dev libpq5 libruby3.2 libssl-dev libssl3t64 libsystemd-shared libsystemd0 libudev1
  linux-libc-dev openssl php8.3-bcmath php8.3-cgi php8.3-cli php8.3-common php8.3-dev php8.3-mbstring php8.3-mysql php8.3-opcache
  php8.3-readline php8.3-xml qemu-user qemu-user-static ruby3.2 ruby3.2-dev systemd systemd-dev systemd-sysv tzdata udev
59 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
Need to get 894 MB of archives.
```

Step 2: Install Required Dependencies

```
g2022_aditi_taksale@cloudshell:~$ sudo apt install -y jq curl git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
jq is already the newest version (1.7.1-3ubuntu0.24.04.1).
curl is already the newest version (8.5.0-2ubuntu10.8).
git is already the newest version (1:2.43.0-1ubuntu7.3).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
```

Then verify everything is present:

```
node -v
```

```
npm -v
```

```
git --version
```

```
docker --version
```

```
docker ps
```

```
g2022_aditi_taksale@cloudshell:~$ node -v
```

```
v24.14.1
```

```
g2022_aditi_taksale@cloudshell:~$ npm -v
```

```
11.12.1
```

```
g2022_aditi_taksale@cloudshell:~$ git --version
```

```
git version 2.43.0
```

```
g2022_aditi_taksale@cloudshell:~$ docker --version
```

```
Docker version 29.4.0, build 9d7ad9f
```

```
g2022_aditi_taksale@cloudshell:~$
```

```
g2022_aditi_taksale@cloudshell:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
g2022	aditi	taksale@cloudshell:~\$				

Step 3: Fix Docker Compose

```
DOCKER_CONFIG=${DOCKER_CONFIG:-$HOME/.docker}
```

```
mkdir -p $DOCKER_CONFIG/cli-plugins
```

```
curl -SL
```

```
https://github.com/docker/compose/releases/download/v2.24.5/docker-compose
```

```
-linux-x86_64 \
```

```
-o $DOCKER_CONFIG/cli-plugins/docker-compose
```

```
chmod +x $DOCKER_CONFIG/cli-plugins/docker-compose
```

```
docker compose version
```

```
sudo mv /usr/bin/docker-compose /usr/bin/docker-compose-old 2>/dev/null ||
```

```
true
```

```
sudo tee /usr/local/bin/docker-compose > /dev/null <<'EOF'
```

```
#!/bin/bash
```

```
exec docker compose "$@"
```

```
EOF
```

```
sudo chmod +x /usr/local/bin/docker-compose
```

```
docker compose version
```

docker-compose version

```
g2022_anuprita_mhapankar@cloudshell:~$ DOCKER_CONFIG=${DOCKER_CONFIG:-$HOME/.docker}
mkdir -p $DOCKER_CONFIG/cli-plugins
curl -SL https://github.com/docker/compose/releases/download/v2.24.5/docker-compose-linux-x86_64 \
-o $DOCKER_CONFIG/cli-plugins/docker-compose
chmod +x $DOCKER_CONFIG/cli-plugins/docker-compose
docker compose version
% Total      % Received % Xferd Average Speed   Time    Time     Time  Current
   0      0     0     0         0      0      0      0      0
100 58.5M  100 58.5M    0     0  79.4M      0  --:--:-- --:--:-- --:--:--   79.4M
Docker Compose version v2.24.5
g2022_anuprita_mhapankar@cloudshell:~$ sudo mv /usr/bin/docker-compose /usr/bin/docker-compose-old 2>/dev/null || true
g2022_anuprita_mhapankar@cloudshell:~$ sudo tee /usr/local/bin/docker-compose > /dev/null <<'EOF'
#!/bin/bash
exec docker compose "$@"
EOF
sudo chmod +x /usr/local/bin/docker-compose
g2022_anuprita_mhapankar@cloudshell:~$ docker compose version
docker-compose version
Docker Compose version v2.24.5
Docker Compose version v2.24.5
g2022_anuprita_mhapankar@cloudshell:~$
```

Step 4: Clone Fabric Samples

```
cd $HOME
git clone https://github.com/hyperledger/fabric-samples
cd fabric-samples
```

```
g2022_anuprita_mhapankar@cloudshell:~$ cd $HOME
g2022_anuprita_mhapankar@cloudshell:~$ git clone https://github.com/hyperledger/fabric-samples
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15618, done.
remote: Counting objects: 100% (238/238), done.
remote: Compressing objects: 100% (129/129), done.
remote: Total 15618 (delta 161), reused 109 (delta 109), pack-reused 15380 (from 3)
Receiving objects: 100% (15618/15618), 24.07 MiB | 17.13 MiB/s, done.
Resolving deltas: 100% (8809/8809), done.
g2022_anuprita_mhapankar@cloudshell:~$ cd fabric-samples
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 5: Install Fabric Binaries and Docker Images

```
curl -sSLO
https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-
fabric.sh
chmod +x install-fabric.sh
./install-fabric.sh docker samples binary
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ chmod +x install-fabric.sh
./install-fabric.sh docker samples binary
```

Clone hyperledger/fabric-samples repo

==> Already in fabric-samples repo

fabric-samples v2.5.15 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric binaries

==> Downloading version 2.5.15 platform specific fabric binaries

==> Downloading: https://github.com/hyperledger/fabric/releases/download/v2.5.15/hyperledger-fabric-linux-amd64-2.5.15.tar.gz

==> Will unpack to: /home/g2022_anuprita_mhapankar/fabric-samples

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
0	0	0	0	0	0	--:--:--	0
100	120M	100	120M	0	0	0:00:05	24.4M

==> Done.

==> Downloading version 1.5.17 platform specific fabric-ca-client binary

==> Downloading: https://github.com/hyperledger/fabric-ca/releases/download/v1.5.17/hyperledger-fabric-ca-linux-amd64-1.5.17.tar.gz

==> Will unpack to: /home/g2022_anuprita_mhapankar/fabric-samples

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
0	0	0	0	0	0	--:--:--	0
100	31.6M	100	31.6M	0	0	0:00:01	31.6M

==> Done.

Pull Hyperledger Fabric docker images

FABRIC_IMAGES: peer orderer ccenv baseos

==> Pulling fabric Images

==> ghcr.io/hyperledger/fabric-peer:2.5.15

====> ghcr.io/hyperledger/fabric-ca:1.5.17

1.5.17: Pulling from hyperledger/fabric-ca

18afb8c8d2a4: Pull complete

6f4ebca3e823: Pull complete

b9649d3f054a: Pull complete

0b966e606d9a: Pull complete

c36bacd85367: Download complete

Digest: sha256:453a0a727e82c4960b797e379adc231e853ee23ae7526db53afef77b5074117e

Status: Downloaded newer image for ghcr.io/hyperledger/fabric-ca:1.5.17

ghcr.io/hyperledger/fabric-ca:1.5.17

==> List out hyperledger images

WARNING: This output is designed for human readability. For machine-readable output, please use --format.

ghcr.io/hyperledger/fabric-baseos:2.5.15	654bd4554bf9	250MB	80.4MB
ghcr.io/hyperledger/fabric-ca:1.5.17	453a0a727e82	381MB	123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.15	86dded7eda92	1GB	255MB
ghcr.io/hyperledger/fabric-orderer:2.5.15	9972c209be47	178MB	49.5MB
ghcr.io/hyperledger/fabric-peer:2.5.15	3e25adc31a75	230MB	66.2MB
hyperledger/fabric-baseos:2.5	654bd4554bf9	250MB	80.4MB
hyperledger/fabric-baseos:2.5.15	654bd4554bf9	250MB	80.4MB
hyperledger/fabric-baseos:latest	654bd4554bf9	250MB	80.4MB
hyperledger/fabric-ca:1.5	453a0a727e82	381MB	123MB
hyperledger/fabric-ca:1.5.17	453a0a727e82	381MB	123MB
hyperledger/fabric-ca:latest	453a0a727e82	381MB	123MB
hyperledger/fabric-ccenv:2.5	86dded7eda92	1GB	255MB
hyperledger/fabric-ccenv:2.5.15	86dded7eda92	1GB	255MB
hyperledger/fabric-ccenv:latest	86dded7eda92	1GB	255MB
hyperledger/fabric-orderer:2.5	9972c209be47	178MB	49.5MB
hyperledger/fabric-orderer:2.5.15	9972c209be47	178MB	49.5MB
hyperledger/fabric-orderer:latest	9972c209be47	178MB	49.5MB
hyperledger/fabric-peer:2.5	3e25adc31a75	230MB	66.2MB
hyperledger/fabric-peer:2.5.15	3e25adc31a75	230MB	66.2MB
hyperledger/fabric-peer:latest	3e25adc31a75	230MB	66.2MB

g2022_anuprita_mhapankar@cloudshell:~/fabric-samples\$

Step 6: Verify Docker images

docker images | grep fabric

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ docker images | grep fabric
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
ghcr.io/hyperledger/fabric-baseos:2.5.15      654bd4554bf9      250MB      80.4MB
ghcr.io/hyperledger/fabric-ca:1.5.17          453a0a727e82      381MB      123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.15       86dded7eda92      1GB        255MB
ghcr.io/hyperledger/fabric-orderer:2.5.15     9972c209be47      178MB      49.5MB
ghcr.io/hyperledger/fabric-peer:2.5.15        3e25adc31a75      230MB      66.2MB
hyperledger/fabric-baseos:2.5                 654bd4554bf9      250MB      80.4MB
hyperledger/fabric-baseos:2.5.15              654bd4554bf9      250MB      80.4MB
hyperledger/fabric-baseos:latest              654bd4554bf9      250MB      80.4MB
hyperledger/fabric-ca:1.5                     453a0a727e82      381MB      123MB
hyperledger/fabric-ca:1.5.17                  453a0a727e82      381MB      123MB
hyperledger/fabric-ca:latest                  453a0a727e82      381MB      123MB
hyperledger/fabric-ccenv:2.5                  86dded7eda92      1GB        255MB
hyperledger/fabric-ccenv:2.5.15               86dded7eda92      1GB        255MB
hyperledger/fabric-ccenv:latest               86dded7eda92      1GB        255MB
hyperledger/fabric-orderer:2.5                9972c209be47      178MB      49.5MB
hyperledger/fabric-orderer:2.5.15             9972c209be47      178MB      49.5MB
hyperledger/fabric-orderer:latest             9972c209be47      178MB      49.5MB
hyperledger/fabric-peer:2.5                   3e25adc31a75      230MB      66.2MB
hyperledger/fabric-peer:2.5.15                3e25adc31a75      230MB      66.2MB
hyperledger/fabric-peer:latest                3e25adc31a75      230MB      66.2MB
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 7: Set Environment Variables

export PATH=\$PATH:\$HOME/fabric-samples/bin

export FABRIC_CFG_PATH=\$HOME/fabric-samples/config

peer version

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ export PATH=$PATH:$HOME/fabric-samples/bin
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ peer version
peer:
  Version: v2.5.15
  Commit SHA: 83c7930
  Go version: go1.26.0
  OS/Arch: linux/amd64
  Chaincode:
    Base Docker Label: org.hyperledger.fabric
    Docker Namespace: hyperledger
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 8: Start the Network and Create Channel

cd \$HOME/fabric-samples/test-network

./network.sh up createChannel

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 9: Deploy Chaincode

```
./network.sh deployCC \
```

```
-ccn basic \
```

```
-ccp ../asset-transfer-basic/chaincode-javascript \
```

```
-ccl javascript
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ cd $HOME/fabric-samples/test-network

./network.sh deployCC \
-ccn basic \
-ccp ../asset-transfer-basic/chaincode-javascript \
-ccl javascript
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ '[' false = true ']'
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-basic/chaincode-javascript --lang node --label basic_1.0
+ res=0
Chaincode is packaged
Installing chaincode on peer0.org1...
Using organization 1
```

```
Checking the commit readiness of the chaincode definition successful on peer0.org2 on channel 'mychannel'
Using organization 1
Using organization 2
+ peer lifecycle chaincode commit -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile /home/g2022_anuprita_mhapankar/fabric-samples/test-network/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --channelID mychannel --name basic --peerAddresses localhost:7051 --tlsRootCertFiles /home/g2022_anuprita_mhapankar/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem --peerAddresses localhost:9051 --tlsRootCertFiles /home/g2022_anuprita_mhapankar/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem --version 1
sequence 1
+ res=0
2026-04-09 15:02:00.641 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [d578fad7d5ea9fbbab1c00b0f3bb78e4c57add2a11c8387b9bed0a90efd9559c] committed with status (VALID)
alhost:7051
2026-04-09 15:02:00.646 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [d578fad7d5ea9fbbab1c00b0f3bb78e4c57add2a11c8387b9bed0a90efd9559c] committed with status (VALID)
alhost:9051
Chaincode definition committed on channel 'mychannel'
Using organization 1
Querying chaincode definition on peer0.org1 on channel 'mychannel'...
Attempting to Query committed status on peer0.org1, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Querying chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to Query committed status on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org2 on channel 'mychannel'
Chaincode initialization is not required
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 10: Set Peer Environment Variables

```
export FABRIC_CFG_PATH=$HOME/fabric-samples/config
```

```
export PATH=$PATH:$HOME/fabric-samples/bin
```

```
export CORE_PEER_TLS_ENABLED=true
```

```
export CORE_PEER_LOCALMSPID="Org1MSP"
```

```
export
```

```
CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organization
s/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.c
rt
```



```
export
CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
export PATH=$PATH:$HOME/fabric-samples/bin
export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
export CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 11: Initialize Ledger

```
peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
  -c '{"function": "InitLedger", "Args": []}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
-n mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
-c '{"function": "InitLedger", "Args": []}'
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 12: Query All Assets

```
peer chaincode query -C mychannel -n basic -c '{"Args": ["GetAllAssets"]}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args": ["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}, {"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"}, {"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"}, {"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"}, {"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}, {"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 13: Read a Specific Asset

```
peer chaincode query -C mychannel -n basic -c
 '{"Args": ["ReadAsset", "asset5"]}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args": ["ReadAsset", "asset5"]}'
{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```


Step 14: Create a New Asset

```
peer chaincode invoke \  
  -o localhost:7050 \  
  --ordererTLSHostnameOverride orderer.example.com \  
  --tls \  
  --cafile  
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \  
  -C mychannel \  
  -n basic \  
  --peerAddresses localhost:7051 \  
  --tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \  
  --peerAddresses localhost:9051 \  
  --tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \  
  -C  
'{"function": "CreateAsset", "Args": ["asset7", "blue", "10", "Anuprita", "500"]}'  
,
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \  
-c '{"function": "CreateAsset", "Args": ["asset7", "blue", "10", "Anuprita", "500"]}'  
$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem -C mychannel -n basic --peerAddresses localhost:7051 --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt --peerAddresses localhost:9051 --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt -c '{"function": "CreateAsset", "Args": ["asset7", "blue", "10", "Anuprita", "500"]}'  
Chaincode invoke successful. result: status:200 payload:"{\"ID\": \"asset7\", \"Color\": \"blue\", \"Size\": 10, \"Owner\": \"Anuprita\", \"AppraisedValue\": 500}"  
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 15: Transfer Asset

```
peer chaincode invoke \  
  -o localhost:7050 \  
  --ordererTLSHostnameOverride orderer.example.com \  
  --tls \  
  --cafile  
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \  
  -C mychannel \  
  -n basic \  
  --peerAddresses localhost:7051 \  
  --tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \  
  --peerAddresses localhost:9051 \  
  --tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt
```

```
--peerAddresses localhost:9051 \
--tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
-c '{"function": "TransferAsset", "Args": ["asset7", "Sneha"]}'
```

```
q2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
-c '{"function": "TransferAsset", "Args": ["asset7", "Sneha"]}'
2024-04-09 15:04:35.967 UTC [INFO] [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"Anuprita"
q2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 16: Stop the Network

```
cd $HOME/fabric-samples/test-network
./network.sh down
```

```
q2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ cd $HOME/fabric-samples/test-network
./network.sh down
Stopping network
Using docker and docker-compose
project name can't be empty. Use '--project-name' to set a valid name
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
c72f6a9969f0
a811e27601bb
5f5198e286ca
3bcfb908afe1
8d2fc8e74375
Removing generated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbc664c87b24c035bee15cbcc7f59b629-c68f8ecc2a0ff0cc1497615f5e33a8bfa908e0fd3555f864cfa1398b52d341c
b:latest
Deleted: sha256:fa120af33c4db851b5dded871dd45c39e7e3567eb3d753fccfca850a32284087
Untagged: dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbc664c87b24c035bee15cbcc7f59b629-d3cf9b1f4b9747cc8433ac0e74bfaeade1cf0b97f5c1163835f604aa885d60d
c:latest
Deleted: sha256:edc7441a9a8c48b025dba4a79878631a900a26be317335ae9c9b2512c869f810
q2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

CONCLUSION:

The Hyperledger Fabric network was successfully implemented and chaincode was deployed. Asset operations such as creation, querying, and transfer were performed successfully. This experiment demonstrated secure and efficient asset management using a permissioned blockchain system.